

HOW LPS SHAPES LPTB₂FG DYNAMICS IN A NATIVE-LIKE INNER MEMBRANE

Dario DE VECCHIS¹, Cedric LAGURI¹, Franck FIESCHI¹

¹ *Univ. Grenoble Alpes, CNRS, CEA, Institute of structural biology, Grenoble, France.*

(corresponding author: dario.de-vecchis@ibs.fr)

Keywords: lipopolysaccharide transport ; multiscale molecular dynamics ; lipid-protein interactions ; membrane protein dynamics ; lipid organization.

ABSTRACT

Gram-negative bacteria rely on lipopolysaccharide (LPS) to build the outer-membrane permeability barrier that contributes to intrinsic antimicrobial resistance. The ABC transporter LptB₂FG extracts LPS from the inner membrane, but how substrate recognition shapes transporter dynamics remains unclear because most current models come from detergent or nanodisc-reconstituted systems rather than native-like membranes.

Here, we use multiscale molecular dynamics simulations, combining coarse-grained and atomistic approaches, to examine how LPS modulates LptB₂FG dynamics in a native-like Gram-negative inner membrane. Our initial simulations suggest that LPS can contact the transporter through its sugar-rich headgroup before acyl-chain entry into the cavity, with extended moieties such as the O-antigen reaching beyond the transporter boundaries into the periplasm. We therefore hypothesize that distinct LPS regions influence early recognition, lipid accessibility at the transporter mouth, and local membrane organization during the first steps of extraction.

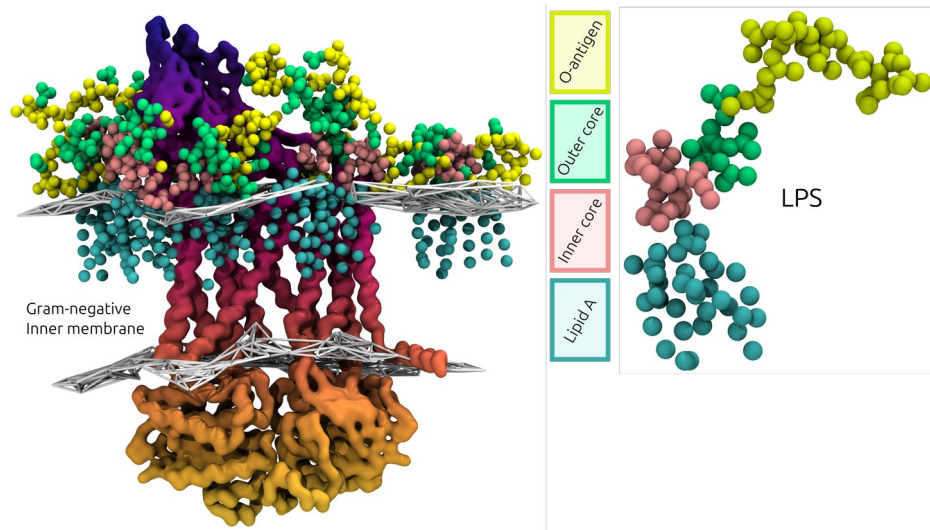


Fig. *LptB₂FG* transporter embedded in a model Gram-negative inner membrane in presence of its substrate lipopolysaccharide (LPS, on the right).